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End-to-end verification workflow

FastAPI backend · React/Vite frontend · local open-source AI models · no raw image storage

VERIFICATION FLOW — EVERY STEP MUST PASS

1 · Start session

BROWSER + API

- User enters a real **user_id**
- **POST /api/verifications**
- Assigns 3 random face actions
- Assigns 3 hand gestures (max 1 tricky)

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2 · Document proofing

BACKEND ANALYSIS

- Passport / Lao ID upload or camera
- Quality: sharpness, glare, contrast
- OCR + TD3 MRZ check digits (hard fail)
- Recapture, tamper + model ensemble
- Portrait face embedding kept in session

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3 · Active liveness

BROWSER + BACKEND

- Blink / turn head / open mouth
- FaceMesh detects the action live
- 3-frame evidence burst to backend
- Backend rejects screen/photo replay

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4 · Hand gesture

BROWSER

- 3 gestures from 23-gesture catalog
- MediaPipe Hands tracks the hand
- Hold gesture inside the moving circle
- Progress ring completes each pass

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5 · Selfie match + PAD

BACKEND ANALYSIS

- 10-frame live burst
- Face-size gate: fill the yellow circle
- facenox PAD + replay heuristics
- Natural motion / lighting checks
- SFace match vs document portrait ≥ 0.68

Decision engine

- Recomputed after every step
- Any hard-fail signal → **rejected**
- All checks passed → **passed**
- Otherwise → **pending**
- Explainable reason codes + scores

PASSED

All checks green: document genuine, person live, gestures done, selfie matches the document portrait. Unlocks Face ID enrollment below.

REJECTED

Hard fail: unreadable/expired document, invalid MRZ, replay or spoof evidence, failed challenge, or face match below threshold. User gets an actionable message and can retry the step.

PENDING

Steps remain. Reason codes such as PASSIVE_LIVENESS_REQUIRED show exactly what is still missing.

AFTER A PASSED VERIFICATION — RETURNING USER

Enroll Face ID

OPTIONAL, AFTER PASS

- **POST .../enroll-face**
- One active Face ID per user_id
- One profile per passport number
- SQLAlchemy: SQLite / PostgreSQL

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Returning face login

ANY LATER VISIT

- **POST /api/face-login**
- Passive liveness first, then match
- SFace vs enrolled templates
- Returns the verified profile

→

Face Pay demo

PAYMENT-STYLE ACTION

- Transfer confirmed by Face ID
- Shows bank-style authorization
- Uses the same login pipeline

Privacy: raw passport and selfie images are analyzed in memory and never stored. Only face templates (after consented enrollment), scores, reason codes, and timestamps persist. Thresholds are tunable via LALIGENCE_* environment variables.